

Woojae Shin

Ph.D. Student in Mechanical Engineering

Korea Advanced Institute of Science and Technology (KAIST)

sindream.github.io

ORCID 0000-0002-6155-3712

Born June 26, 1998

Profile

Ph.D. student at KAIST researching reinforcement learning for autonomous robotic systems, particularly unmanned aerial vehicles. Research interests also include robotic perception, state estimation, and multi-robot systems.

Education

2025 - Present	Ph.D. Student in Mechanical Engineering Korea Advanced Institute of Science and Technology (KAIST)
2025	M.S. in Mechanical Engineering Ulsan National Institute of Science and Technology (UNIST)
2023	B.S. in Mechanical Engineering Ulsan National Institute of Science and Technology (UNIST)

Publications

2026	Vision-Based Autonomous Drone Landing on Moving Platforms With Uncertain Motion via Deep Reinforcement Learning [DOI] Woojae Shin, Minwoo Kim, Taewook Park, Geunsik Bae, Seunghwan Kim, and Hyondong Oh IEEE Robotics and Automation Letters , 11(5), 5542-5549
2026	Real-Time Communication Relay Planning With a Low-Complexity Network Quality Prediction Model in Dynamic Indoor Missions [DOI] Jaemin Seo, Jongyun Kim, Seunghwan Kim, Changseung Kim, Woojae Shin , and Hyondong Oh IEEE Robotics and Automation Letters , 11(1), 322-329
2025	EKF-Based Radar-Inertial Odometry With Online Temporal Calibration [DOI] Changseung Kim, Geunsik Bae, Woojae Shin , Sen Wang, and Hyondong Oh IEEE Robotics and Automation Letters , 10(7), 7230-7237
2024	Autonomous Landing on a Moving Platform Using Vision-Based Deep Reinforcement Learning [DOI] Pawel Ladosz, Meraj Mammadov, Heejung Shin, Woojae Shin , and Hyondong Oh IEEE Robotics and Automation Letters , 9(5), 4575-4582

Research interests

Reinforcement learning / Autonomous UAVs / Robotic perception / State estimation / Multi-robot systems